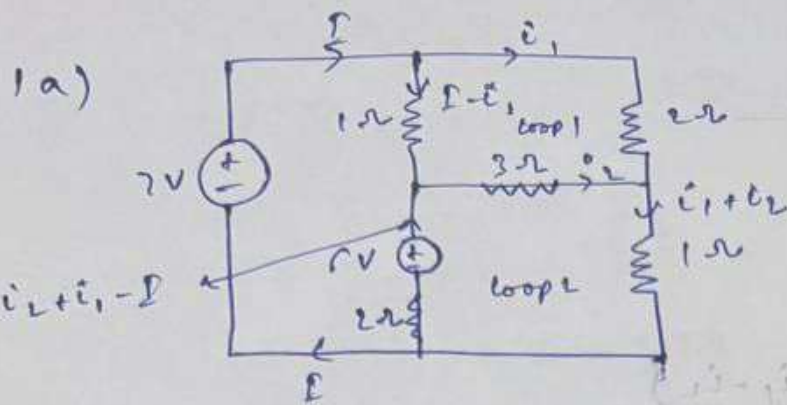




Now marking current in loop, considering KCL equations
applying loop Method to find loop current

KVL in loop ① gives

$$-2i_1 + 3i_2 + 1(I - i_1) = 0$$

$$3i_1 = 3i_2 + I \quad \rightarrow \text{①}$$

Now in loop ② KVL again,

$$-3i_2 - 1(i_1 + i_2) - 2(i_1 + i_2 - I) + 6 = 0$$

$$3i_1 + 6i_2 = 2I + 6 \quad \rightarrow \text{②}$$

Now we have 3 variables in eq ① and ② i.e. $i_1, i_2 + I$
so we need one more equation

$\therefore$ applying KVL in loop

$$-2i_1 - 1(i_1 + i_2) + 7 = 0$$

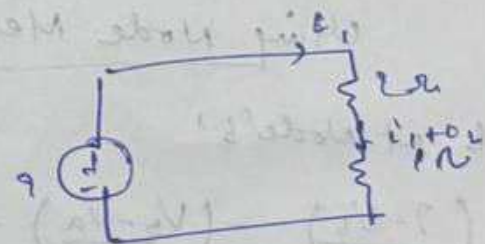
$$3i_1 + i_2 = 7 \quad \rightarrow \text{③}$$

Using ① + ②

$$6i_1 = 6i_2 + 2I$$

$$3i_1 + 6i_2 = 2I + 6$$

$$4i_2 - i_1 = 2 \quad \rightarrow \text{④}$$



using (3) + (4)

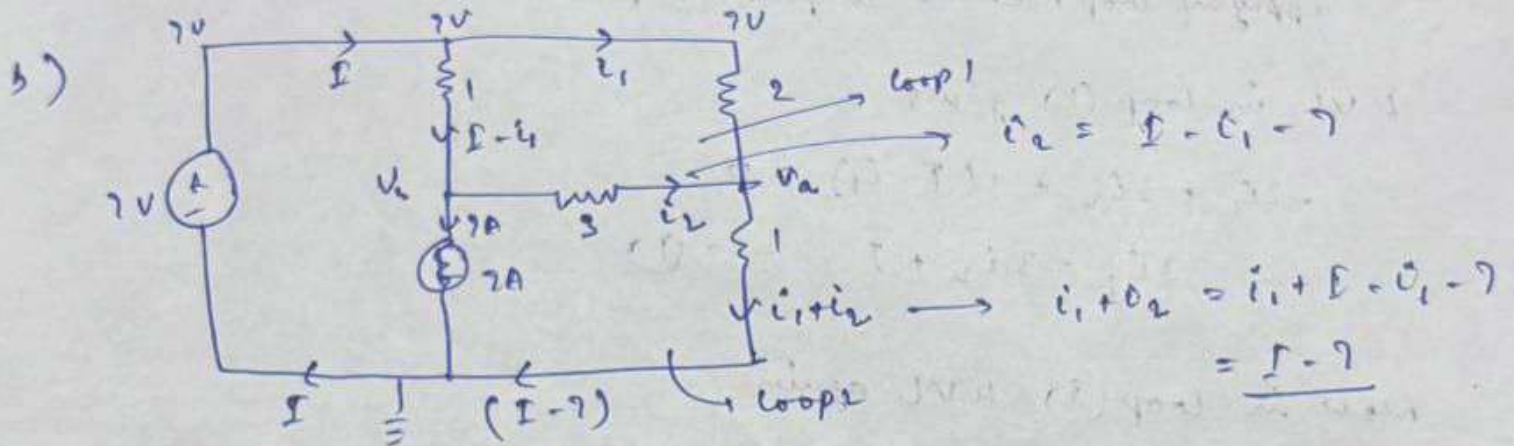
$$\begin{array}{r} 3i_1 + i_2 = 7 \\ 12i_2 - 3i_1 = 6 \\ \hline 13i_2 = 13 \end{array}$$

$$\boxed{i_2 = 1A}$$

$$\Rightarrow \boxed{i_1 = 2A}$$

using (1) $\rightarrow I = 3(i_1 - i_2)$

$$\boxed{I = 3A}$$



We denoted branch currents in branches using KCL as well as in
 $K = 3\Omega$, $i_2 = I - i_1 - 7 \Rightarrow$ by KCL

and $i_1 + i_2 = I - 7$

Let V_a & V_b be placed potentials at Node 'a' & 'b'

Using Node Method:

KCL at Node 'b'

$$\left(\frac{7 - V_b}{1}\right) - \left(\frac{V_b - V_a}{3}\right) - 7 = 0$$

$$\Rightarrow \boxed{4V_b = V_a} \rightarrow (1)$$

KCL at Node 'a'

$$\frac{7 - V_a}{2} + \frac{V_b - V_a}{3} - \frac{V_a - 0}{1} = 0$$

$$11V_a - 2V_b = 21 \rightarrow (1)$$

Using (1) and (2)

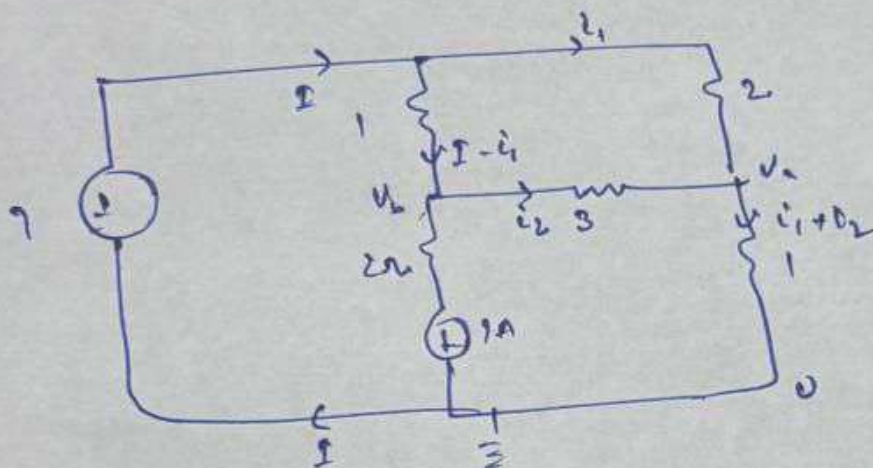
$$V_b = \frac{1}{2} V \quad V_a = 2V$$

Now Currents $i_1 = \frac{7 - 2}{2} = 2.5 A$

$$i_2 = \frac{V_b - V_a}{3} = \frac{\frac{1}{2} - 2}{3} = -0.5 A$$

$$I = i_1 + i_2 + 7 = 2.5 + 0.5 + 7 = 9 A$$

c)



We have denoted currents in loop through KCL

Using Node Method:

KCL at Node 'b':

$$-2 - \left(\frac{V_b - V_a}{3} \right) + \left(\frac{7 - V_b}{1} \right) = 0$$

$$4V_b = V_a \rightarrow (1)$$

KCL at Node 'a':

same eqⁿ from last part

$$11V_a - 2V_b = 21 \rightarrow (2)$$

$\therefore$ No change in current in any loop
 because the resistor we are using/applying is in series with
 constant current source

$\therefore$ No change in current values

$$I = 9 \text{ A}$$

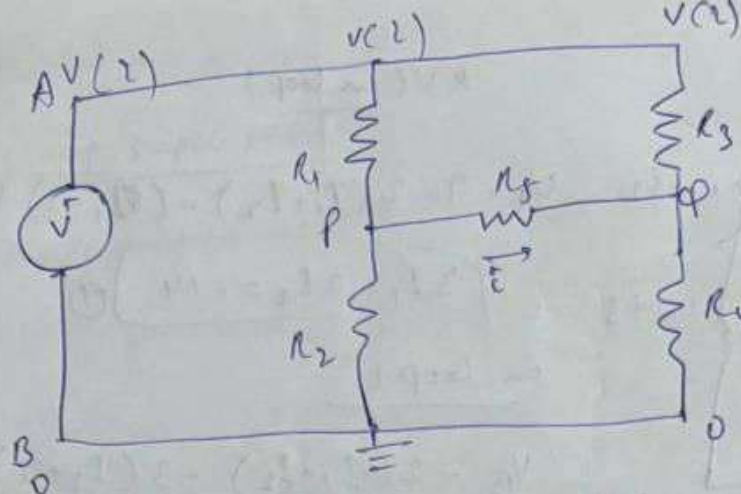
$$I_1 = 2.5 \text{ A}$$

$$I_2 = -0.5 \text{ A}$$

$$AP = 1 + 2.5 + 2.5 = 1 + 5 + 5 = 1$$



2)



$$V = 2V, R_3 = 3\Omega, R_1 = 2\Omega$$

$$R_4 = 2\Omega, R_2 = 4\Omega, R_5 = 1\Omega$$

Assume B at ground voltage

$$i.e. V_B = 0$$

$$\text{and } V_A = 2V$$

We will solve this by nodal method, apply KCL

Applying KCL at node having P volt

$$\frac{P-2}{2} + \frac{P-\Phi}{1} + \frac{P-0}{4} = 0 \quad (1)$$

Applying KCL at node having Φ volt

$$\frac{\Phi-2}{3} + \frac{\Phi-0}{2} + \frac{\Phi-P}{1} = 0 \quad (2)$$

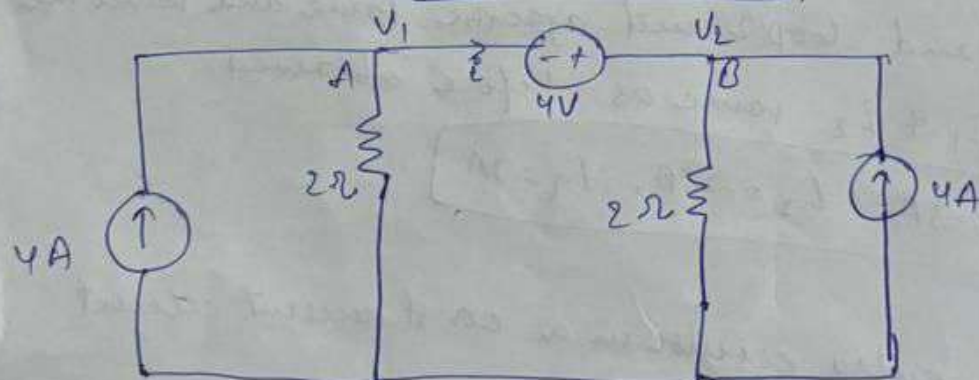
solving (1) and (2)

$$P = \frac{60}{53} = 1.132V$$

$$\Phi = \frac{52}{53} = 0.981V$$

$$\text{now } i = \frac{P-\Phi}{R} = 0.151A$$

3)



To find V_1, V_2 and current i b/w them

In the branch A, B the p.d is const & is equal to 4V

$$V_2 - V_1 = 4V \rightarrow V_2 = V_1 + 4 \quad (1)$$

now applying KCL at node A

$$i + \frac{V_1}{2} = 4 \quad \text{--- (2)}$$

KCL at node B

$$i + 4 + 0 - \frac{V_2}{2} = 0 \quad \text{--- (3)} \quad i + 4 = \frac{V_2}{2}$$

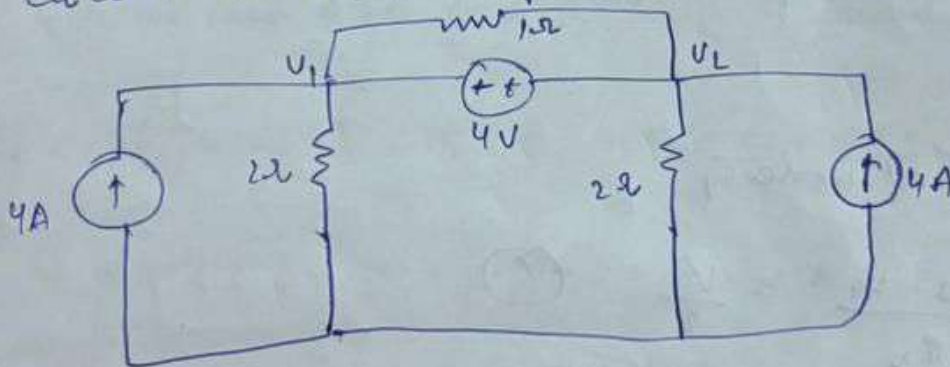
Solve (1) (2) + (3)

$$\frac{V_1}{2} - 4 = 4 - \frac{V_2}{2} \Rightarrow \begin{aligned} V_1 + V_2 &= 16 \\ V_1 - V_2 &= -4 \\ 2V_1 &= 12 \\ \boxed{V_1 = 6V} \end{aligned}$$

Similarly $\boxed{V_2 = 10 \text{ volt}}$
 $i = 1A$

b) Now a 1Ω resistor is added in parallel to branch with nodes V_1 and V_2

Circuit looks something like



again writing eq (1) (2) (3)

eq (1) will still remain same

$$\boxed{V_2 = V_1 + 4}$$

eq (2) $\boxed{\left(\frac{V_1 - V_2}{1} \right) + i + \frac{V_1}{2} = 4}$

↑
extra

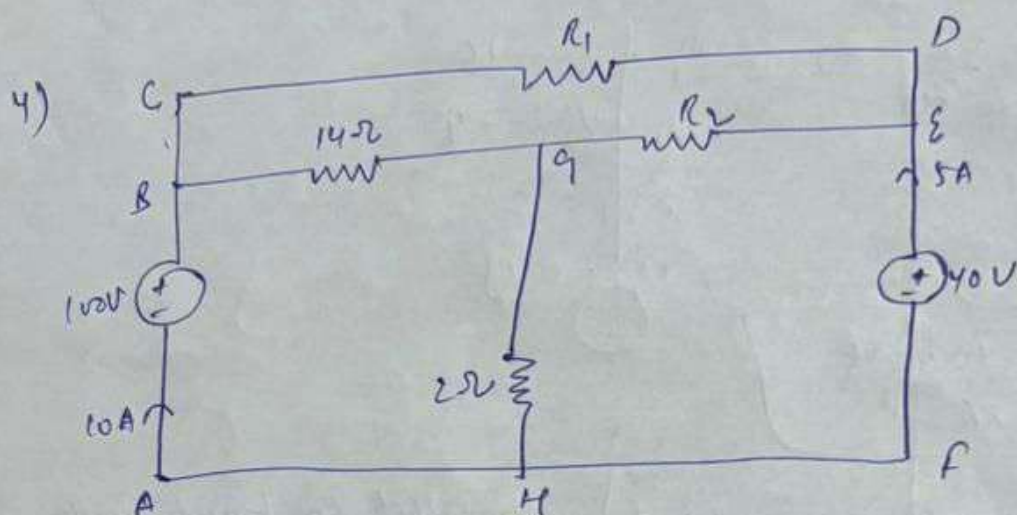
eqn will be

$$V_2 - V_1 + \mathcal{E} + \mathcal{V} = \frac{V_2}{2}$$

solving above eqn we will get

$$\boxed{i = 5A}, \boxed{V_1 = 6V, V_2 = 10V}$$

conclusion: Voltage diff b/w node remained same but current changed



lets take voltage at node $A = 0$

$$V_B = 100 = V_C$$

$$V_H = V_F = 0$$

$$V_E = V_D = 40$$

Applying KCL at node G

$$\frac{100 - V_G}{14} + \frac{40 - V_G}{R_2} = \frac{V_G}{2} \quad (1)$$

Applying KCL at node E

$$i + 5 = \frac{40 - V_G}{R_2} \quad (2)$$

Applying KCL at node B

$$i + \frac{100 - V_G}{14} = 10 \quad (3)$$

Solving above 3 eqns

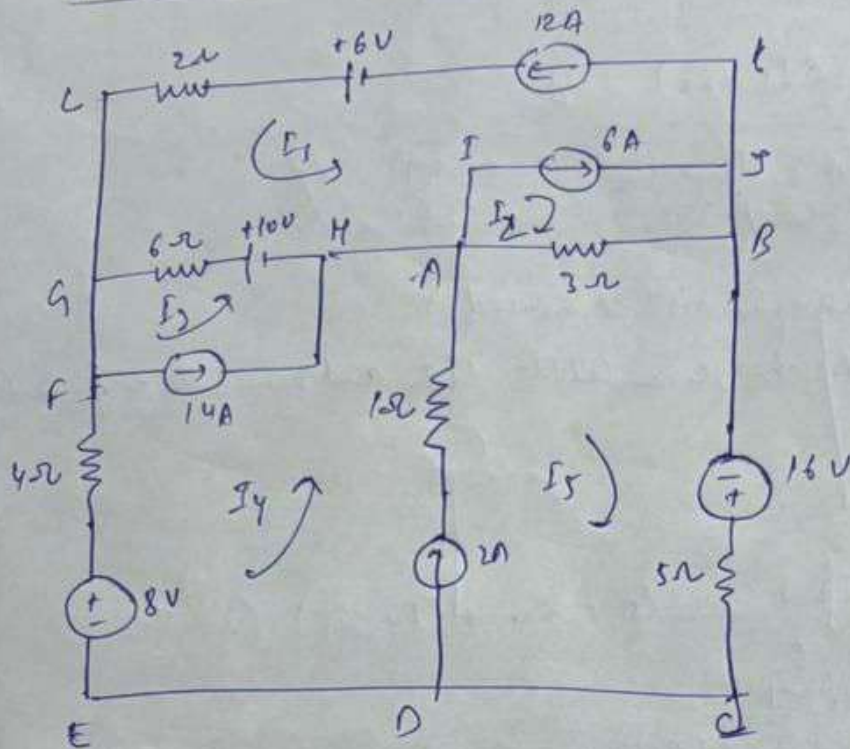
Potential drop across 2Ω resistor $= 2 \times 15 = 30V$

$$V_1 = 30V$$

$$R_2 = 1\Omega$$

$$R_1 = 12\Omega$$

$$\text{and } I = 5A$$



a) Solving for nodal voltages

KVL in loop ABCDEFGH (Let current in this loop I_5)

$$V_A + 3(I_2 - I_5) + 16 - 5(I_5) + 8 + 4(I_4 - I_5) + 6(I_3 - I_5) - 10 = 0$$

Simplifying above eqn $I_1 = 12A$, $I_2 = 6A$, $I_3 = 14A$, $I_4 = 2A$

$$-18I_5 + 124 = 0$$

$$I_5 = 6.89A$$

Ground at C $\Rightarrow V_C = V_D = V_E = 0$

Applying KVL in branch EF

$$V_E + 8 - 4(I_5 - I_4) = V_F$$

on solving we get

$$\begin{aligned} V_F &= -11.56 \\ V_G &= -11.56 \\ V_L &= -11.56 \end{aligned}$$

$\rightarrow$ All are connected

Applying voltage Analysis on $4H$ $9H$

$$V_9 - 6(I_1 - I_3 + I_5) - 10 = V_H$$

$$\boxed{V_H = -50.9} \text{ and } \boxed{V_L = -50.9}$$

$$\boxed{V_A = -50.9}$$

now by $V_A - 3(I_5 - I_2) = V_B$

Upon solving $\boxed{V_B = V_J = V_K = -53.57}$

5) Resistor in $10V$ branch is removed
KVL in loop 5 will change a little bit and hence I_5 only will change

Applying KVL eqn

$$V_A + 3(I_2 - I_5) + 16 - 5I_5 + 8 + 4(I_4 - I_5) - 10 = V_A$$

simplifying

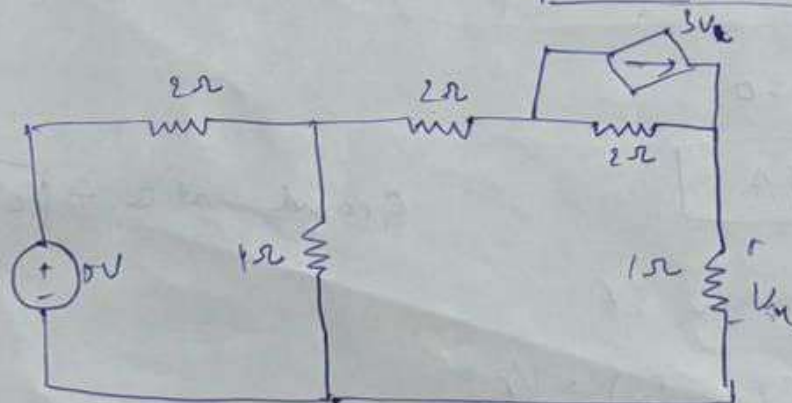
$$12I_5 = 40 \Rightarrow \boxed{I_5 = 3.33A}$$

Voltage analysis on branch EF , GH , AB

$$V_E + 8 - 4(I_5 - I_4) = V_F \Rightarrow \boxed{V_F = V_G = V_L = 2.68V}$$

$$V_9 - 10 = V_H \Rightarrow \boxed{V_H = -7.32 = V_A = V_I}$$

$$V_A - 3(I_5 - I_2) = V_B \Rightarrow \boxed{V_B = V_J = V_K = 0.7V}$$



Applying KVL at node at node A

$$\frac{V_A - 5}{2} + \frac{V_A}{4} + \frac{V_A - V_B}{2} = 0$$

$$\frac{5V_1}{4} - \frac{V_L}{2} = \frac{5}{2} \Rightarrow \boxed{5V_1 - 2V_L = 10} \quad \text{--- (1)}$$

Applying KCL @ B node

$$\frac{V_A - V_B}{2} = 1 + \frac{V_B - V_u}{2}$$

$$V_A - V_B - 5V_u = 0 \quad (2)$$

Applying KCL at node V_u

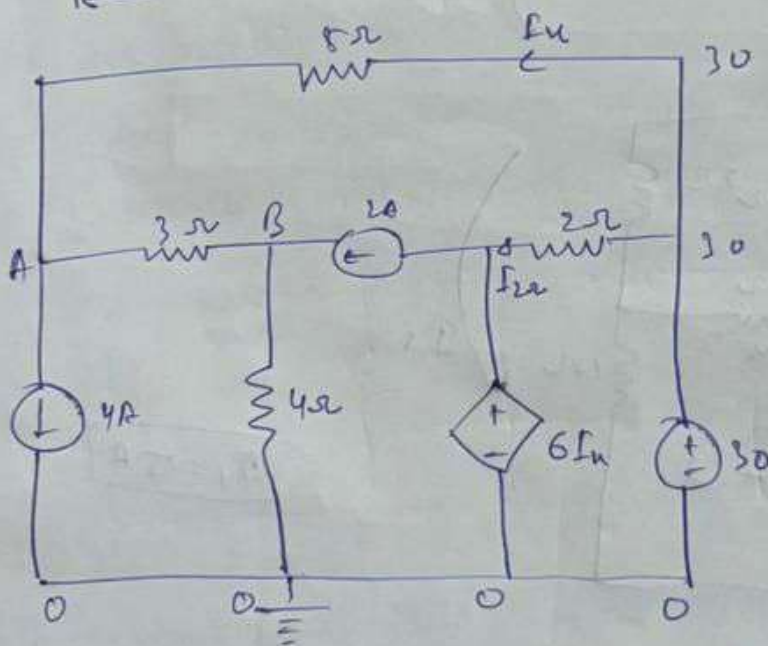
$$3V_u + \frac{V_B - V_u}{2} = V_u$$

$$V_B + 3V_u = 0 \quad (3)$$

Solve (1) (2) (3)

$$\begin{aligned} V_1 &= -10 \text{ volt} \\ V_2 &= -30 \text{ volt} \end{aligned}$$

and we obtain $V_u = 10 \text{ volt}$



Applying KCL at node A

$$\frac{30 - V_A}{5} + \frac{V_B - V_A}{3} = 4$$

$$90 - 3V_A + 5V_B - 5V_A = 60$$

$$8V_A - 5V_B = 30 \quad (1)$$

Applying KCL at node B

$$-\left(\frac{V_B - V_A}{5}\right) + 2 - \frac{V_B}{4} = 0$$

$$4V_A - 9V_B = -24 \quad \text{--- (2)}$$

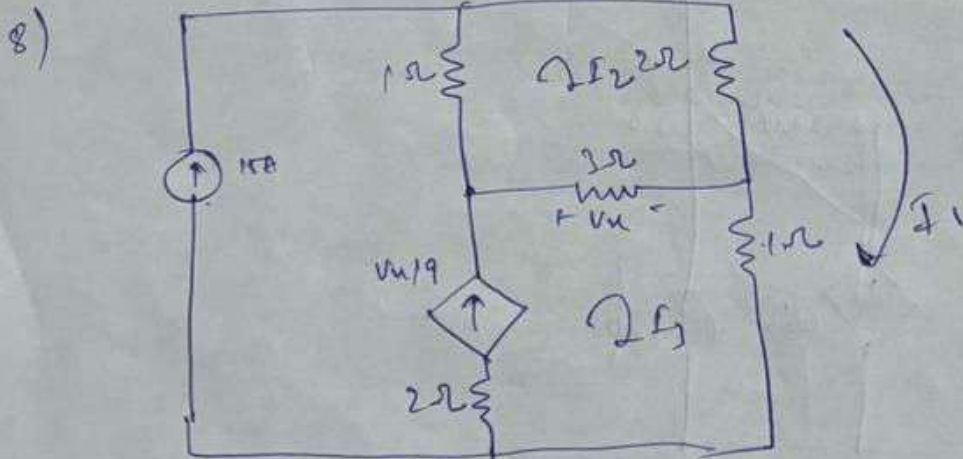
Solving (1) and (2) we get

$$V_A = \frac{55}{6}, \quad V_B = \frac{26}{3}$$

$$V_A = 9.16 \text{ V} \quad V_B = 8.67 \text{ V}$$

$$\text{now } I_u = \frac{30 - V_A}{8} \Rightarrow I_u = 4.17 \text{ A}$$

$$I_{2A} = \frac{30 - 6I_u}{2} = 2.49 \text{ A}$$



$$I_1 = 15 \text{ A}$$

Applying KVL in loop 2

$$V_A - 2(15 + I_2) - 3\left(I_2 - \frac{V_u}{9}\right) - I_2 = 0$$

$$6I_2 - \frac{V_u}{3} = 30 \quad \text{--- (1)}$$

In the outer loop $I = 15 \text{ A}$ (fixed & const)

$$\text{In loop 3} \quad I_3 = \frac{V_u}{9} \quad \text{--- (2)}$$

In branch BC

$$V_u = -3 \left(-\frac{V_u}{9} + I_2 \right)$$

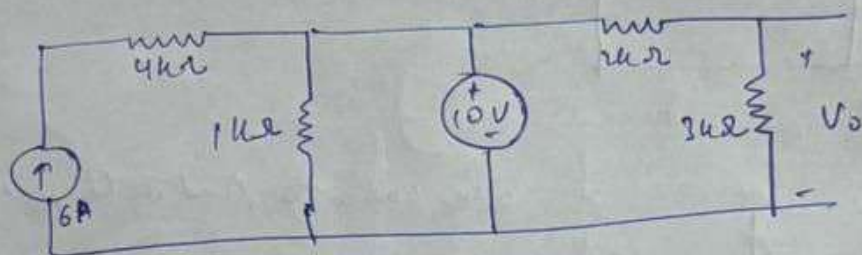
$$2V_u = -9I_2 \quad \text{--- (3)}$$

Solving (1) and (3)

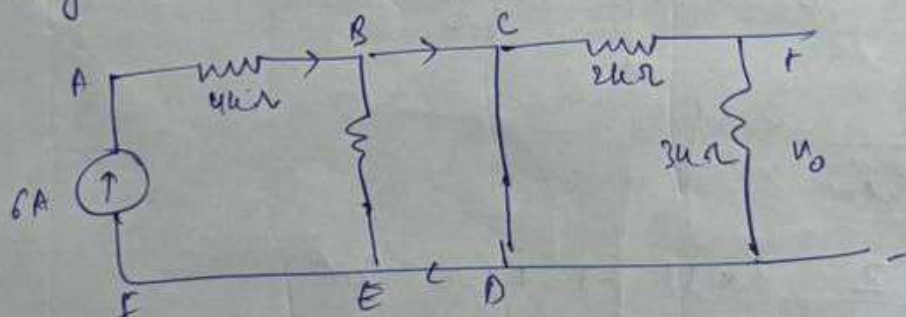
we get
$$\begin{cases} V_u = 18 \text{ volt} \\ I_2 = -4 \text{ A} \end{cases}$$

Using above results
$$I_3 = 2 \text{ A}$$

9)



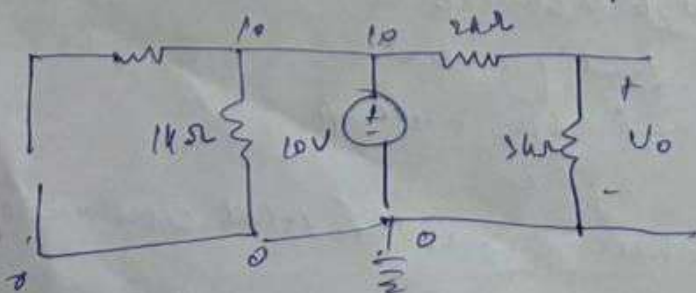
We are required to solve this by superposition method.
Let's first consider the value of V_o due to 6A current source only. We'll short the 10V battery.



All current will flow through ABCDEF

$V_o = 0$ due to 6A current source as all 6A current will flow only through 4kΩ resistor as shown in loop.

Now due to only 10V battery (voltage source)



$$V_o = \left(\frac{3}{3+2} \right) \times 10$$

$$V_o = 6 \text{ V}$$

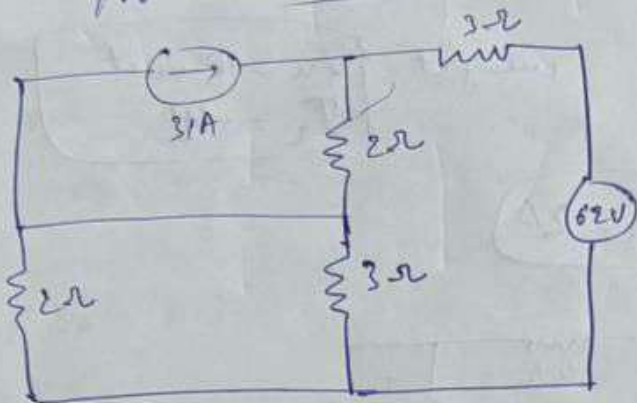
By superposition theorem

$$V_o = V_o(6A) + V_o(10V)$$

$$= 0 + 6$$

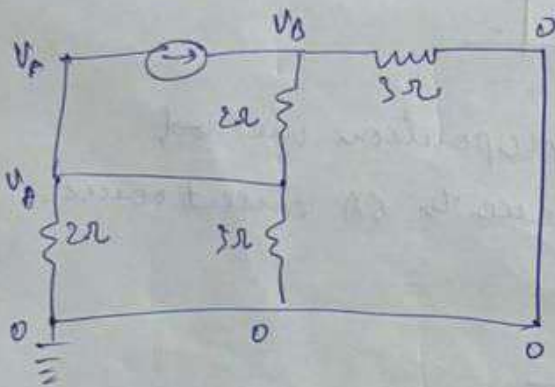
$$V_o = 6 \text{ volts}$$

b)



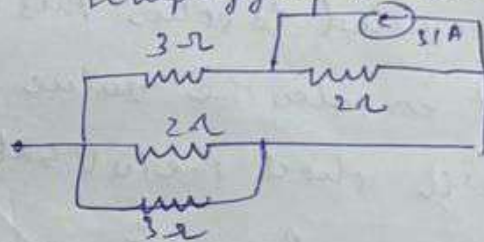
Again solving this by superposition theorem

V_o due to 3A current



We have to find voltage across 3Ω

Simplifying the circuit



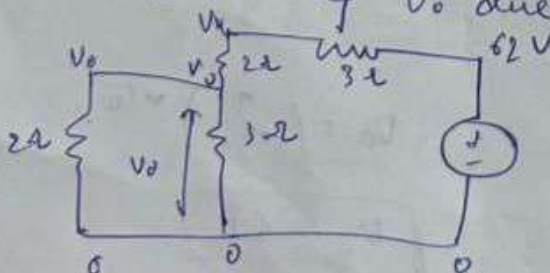
$$i_{3\Omega} = \frac{2}{R_{tot}} \times 3$$

$$= \frac{2}{4.2} \times 3 \Rightarrow i_{3\Omega} = 1.0A$$

$$V_{3\Omega} = -\frac{(1.0 \times 10)}{R_{2||3}} = -12 \text{ volt}$$

$$V_o = -12 \text{ volt due to } 3A \text{ Amp}$$

now calculating V_o due to 62V source



finding current

$$i = \frac{62}{6.2} = 10 \text{ Amp}$$

where 6.2 is the total resistance of given circuit

now P.D across $(2||3)\Omega$ i.e. 1.2Ω will be

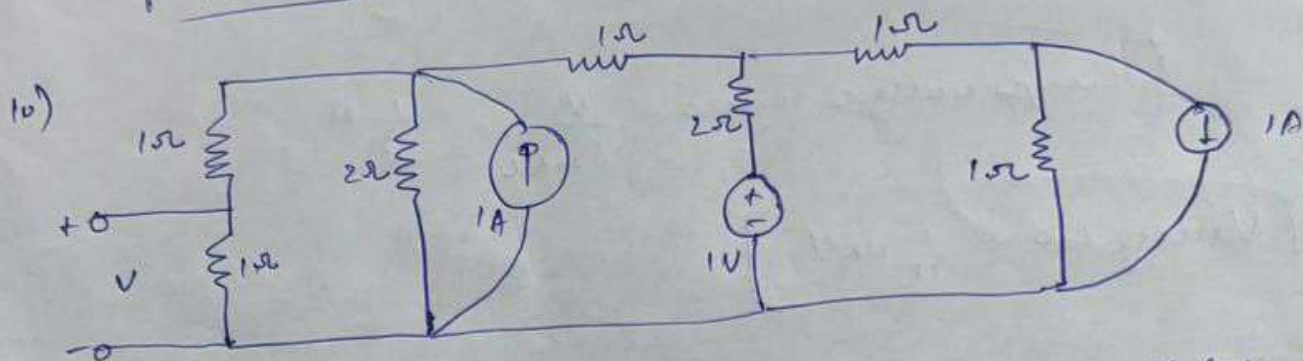
$$V_o = 1.2 \times 10 = 12 \text{ volt}$$

By superposition theorem

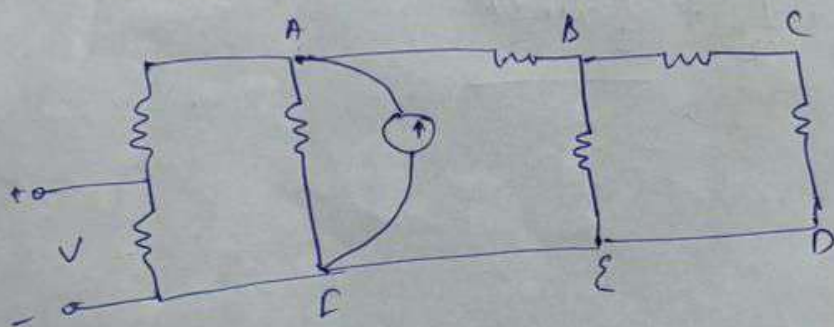
$$V_o = (V_o)_{\text{due to } 6A} + (V_o)_{\text{due to } 2V \text{ cell}}$$

$$= 12 - 12$$

$$V_o = 0$$

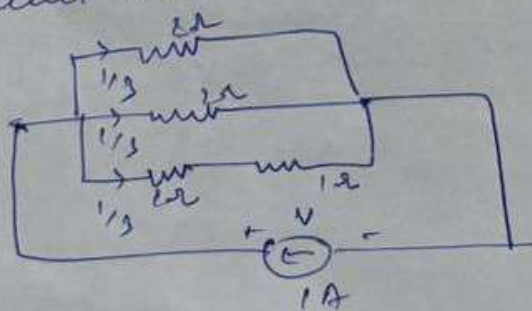


V due to 1A current source (the one parallel to 2Ω)



$$R_{\text{eq of } ABCDEF} = 2\Omega$$

Current will be like

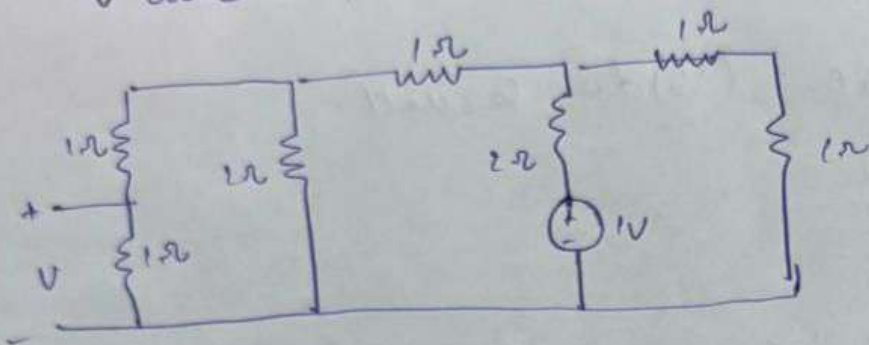


$$V = \frac{1}{3} \text{ volt}$$

11 by V due to 1A current (the one parallel with 1Ω)

$$\text{will be } V = -\frac{1}{12} \text{ volt}$$

V due to 1 volt source



$$R_{eq} = 2\Omega$$

$$i_{\text{through voltage source}} = \frac{V}{R_{\text{total}}} = \frac{1}{3} \text{ A}$$

$$V_{\text{across } 1\Omega} = \frac{1}{12} \text{ volt}$$

By superposition theorem,

$$V = \frac{1}{3} - \frac{1}{12} + \frac{1}{12} = \underline{\underline{\frac{1}{3} \text{ volt}}}$$